

FIGURE · ALGORITHM 2

Client-Side Local Training and Encrypted Gradient Submission

A geometric view of the client-side round procedure — receiving the round model, performing local mini-batch training, averaging and flattening the local gradient, encoding each coordinate with fixed-point scale S , encrypting under Paillier public key pk , and sending the anonymous encrypted submission $M_i^{(t)}$ to the FIP Verifier.

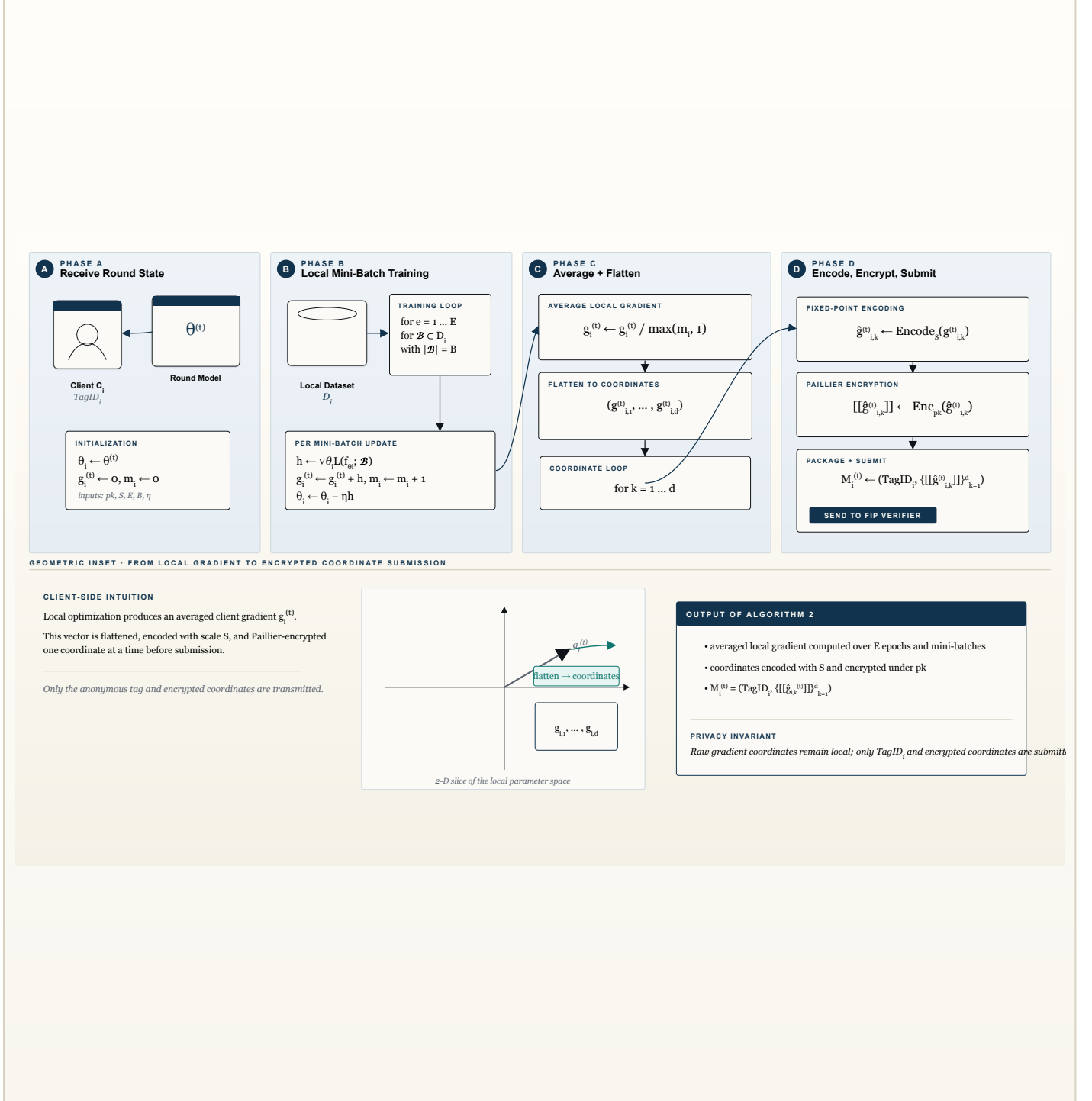


Figure 2. Algorithm 2. Client C_i receives the round model $\theta^{(t)}$, public key pk , fixed-point scale S , and local hyperparameters. It initializes $\theta_i \leftarrow \theta^{(t)}$, performs local mini-batch SGD over E epochs, accumulates and averages the local gradient $g_i^{(t)}$, and then flattens it into coordinates. Each coordinate is encoded as $\hat{g}_{i,k}^{(t)}$ using S and encrypted as $[[\hat{g}_{i,k}^{(t)}]]$ under pk . The transmitted message $M_i^{(t)} = (\text{TagID}_i, \{[[\hat{g}_{i,k}^{(t)}]]\}_{k=1}^d)$ contains only the anonymous tag and encrypted encoded coordinates. The inset summarizes the local descent direction and the flatten–encode–encrypt pipeline.